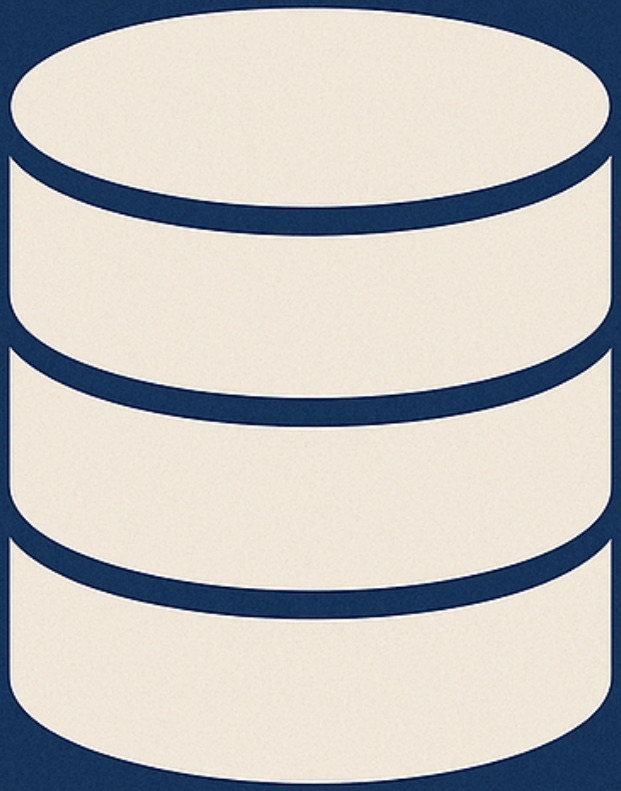
SQL

Class starts
@ 9:10pm



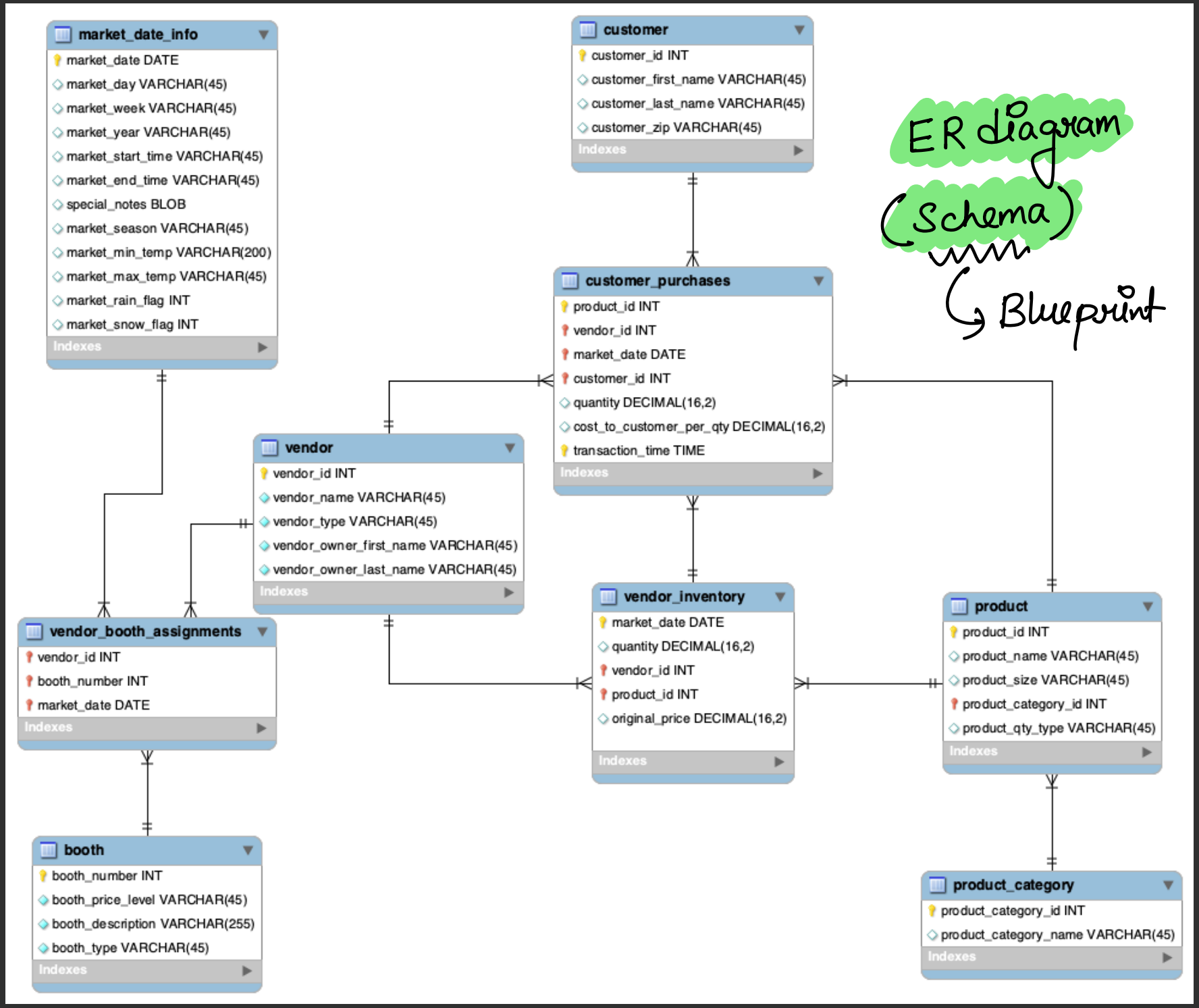


Lecture 5: Applied SQL part 1

#Agenda

1) Data aggregating (grouping of data)

2) Hands on practice problems.


market_date_info
market_date DATE
market_day VARCHAR(45)
market_week VARCHAR(45)
market_year VARCHAR(45)
market_start_time VARCHAR(45)
market_end_time VARCHAR(45)
special_notes BLOB
market_season VARCHAR(45)
market_min_temp VARCHAR(200)
market_max_temp VARCHAR(45)
market_rain_flag INT
market_snow_flag INT
Indexes
customer
customer_id INT
customer_first_name VARCHAR(45)
customer_last_name VARCHAR(45)
customer_zip VARCHAR(45)
Indexes
ER diagram
(Schema)
Blueprint
customer_purchases
product_id INT
vendor_id INT
market_date DATE
customer_id INT
quantity DECIMAL(16,2)
cost_to_customer_per_qty DECIMAL(16,2)
transaction_time TIME
Indexes
vendor
vendor_id INT
vendor_name VARCHAR(45)
vendor_type VARCHAR(45)
vendor_owner_first_name VARCHAR(45)
vendor_owner_last_name VARCHAR(45)
Indexes
vendor_booth_assignments
vendor_id INT
booth_number INT
market_date DATE
Indexes
vendor_inventory
market_date DATE
quantity DECIMAL(16,2)
vendor_id INT
product_id INT
original_price DECIMAL(16,2)
Indexes
product
product_id INT
product_name VARCHAR(45)
product_size VARCHAR(45)
product_category_id INT
product_qty_type VARCHAR(45)
Indexes
booth
booth_number INT
booth_price_level VARCHAR(45)
booth_description VARCHAR(255)
booth_type VARCHAR(45)
Indexes
product_category
product_category_id INT
product_category_name VARCHAR(45)
Indexes

Distinct

ID	Name
1	Aryan
2	Amar
3	Akash
5	Aryan
2	Amar

distinct →

Result

ID	Name
1	Aryan
2	Amar
3	Akash
5	Aryan

Code

```
Select
distinct ID, Name
from
TableNam
```

Data Aggregation

→ group of data

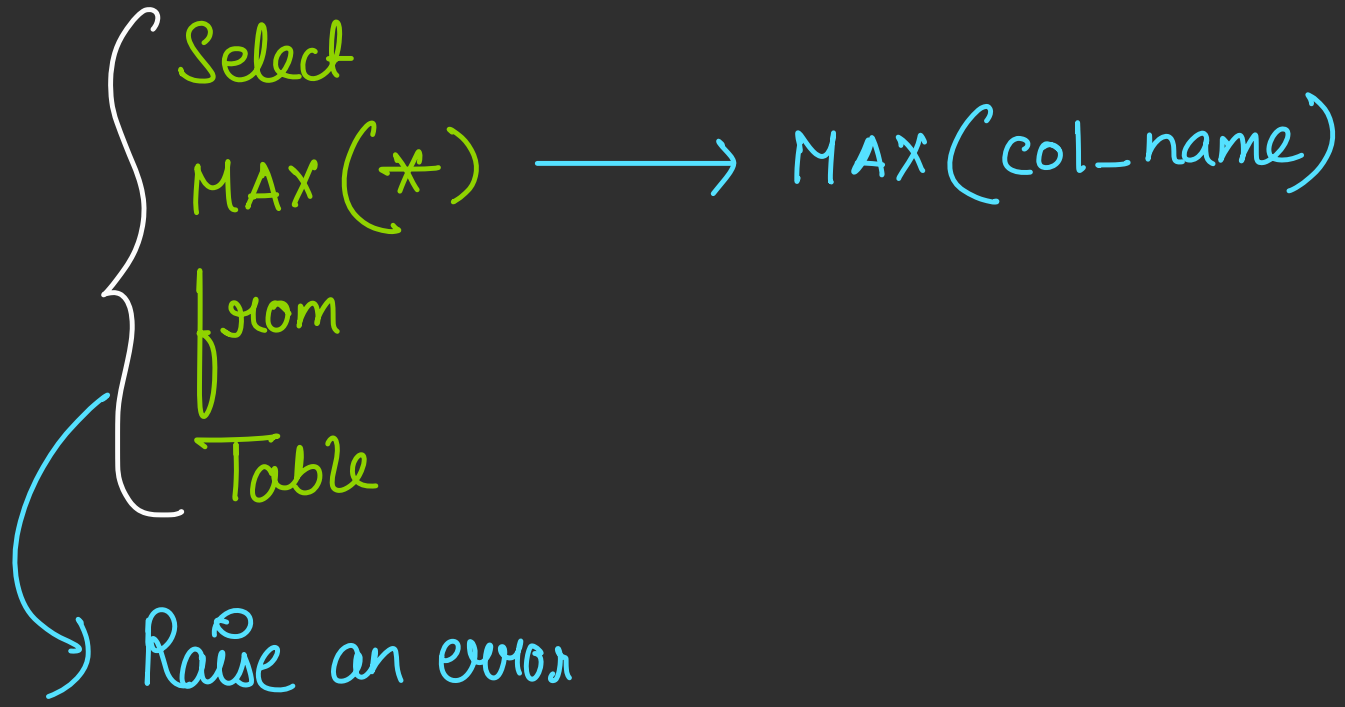
100 rows —Agg→ less than 100 rows

→ Min / Max / Count

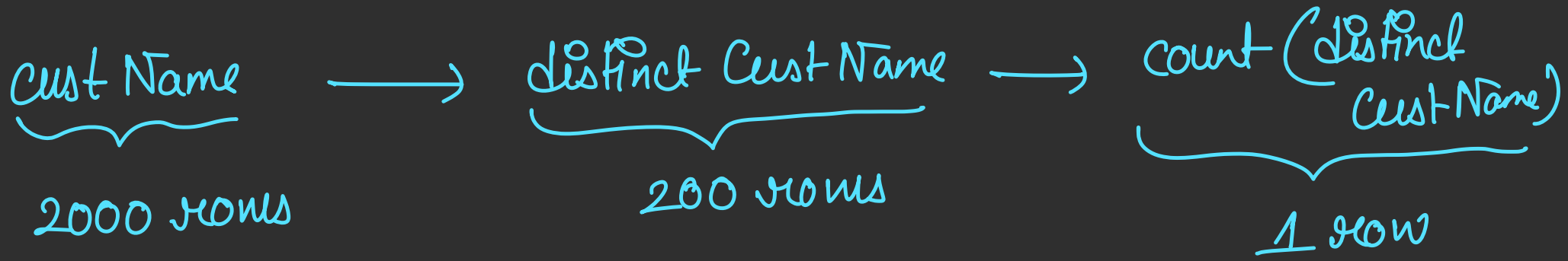
Agg fun^n

```
Select
count(*)
from
Table
```

→ How many records/row is there?


Select
MAX(*)
from
Table
MAX(col_name)
Raise an error

```
Select
count(distinct cust_name)
from
TableA
```



Data Aggregation

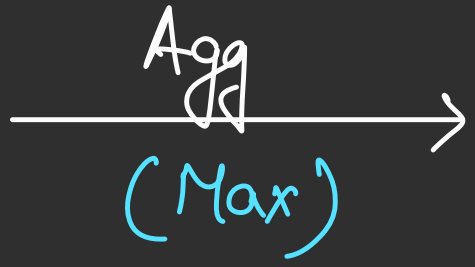
↳ putting things together / grouping.

"Compressing data" is how compression

Table

Col A
13
16
14
12
19

Rows = 5



Col A
19

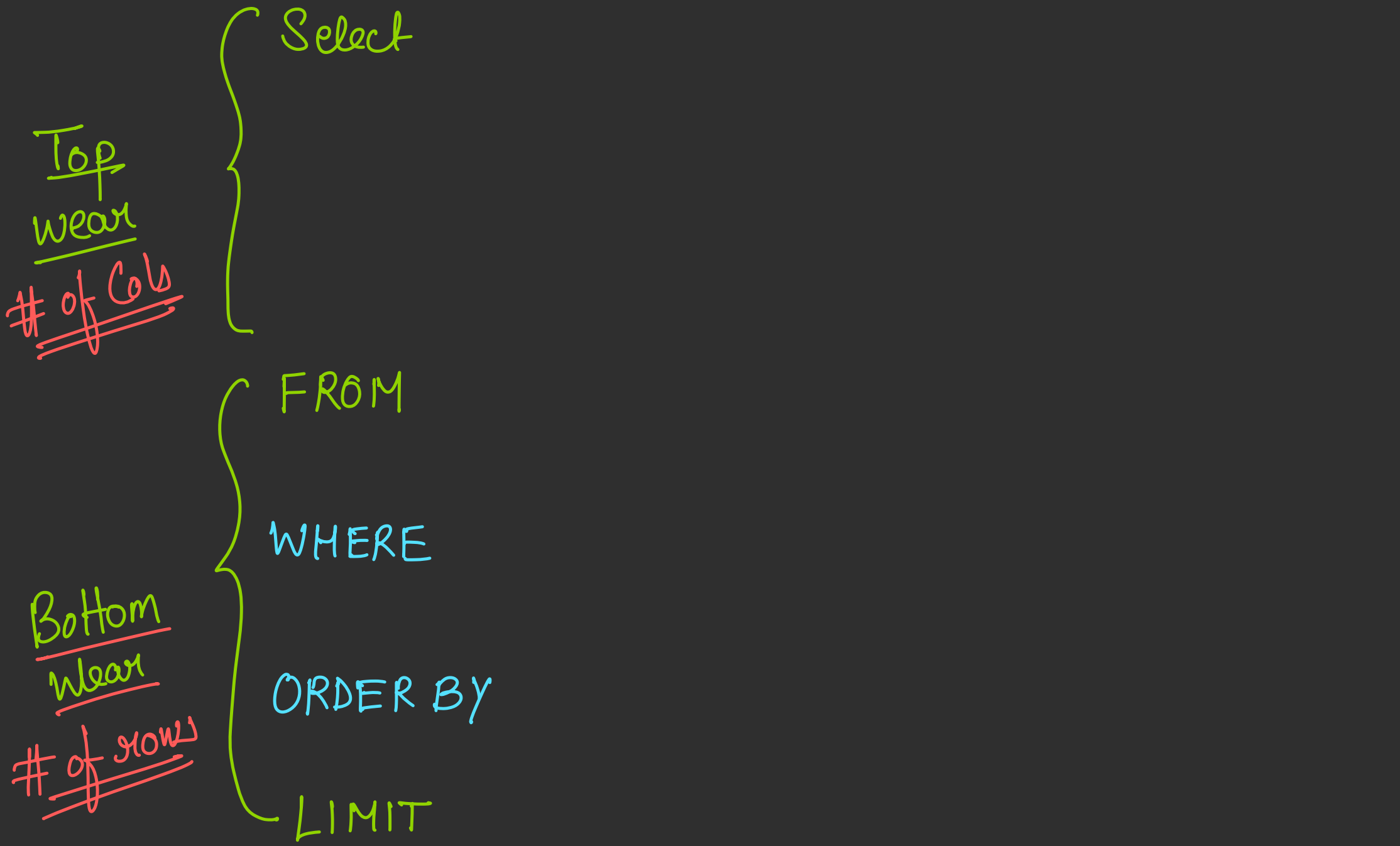
Row = 1

filtering row

(where, ...)

Aggregation

(MAX, MIN, AVG, distinct count)


Select
Top wear
of Cols
FROM
WHERE
ORDER BY
LIMIT
Bottom wear
of rows

booth Type	booth_number
Large	32
Large	42
Small	11

Result ⇒

booth Type	Count booth_number
Large	2
Small	1

Select

*

from

Table

where ⟶ pre filter

group by

having ⟶ post filter (Only & Only used along with group by)

Order by

limit

Order of Execution

1) FROM
2) WHERE
3) GROUP BY
4) Agg fun^{n}
5) SELECT
6) HAVING
7) Order by & Limit